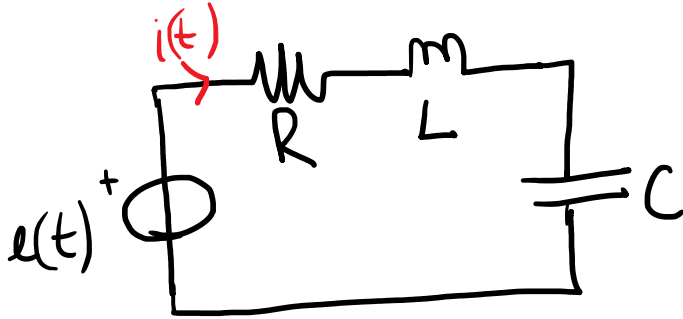


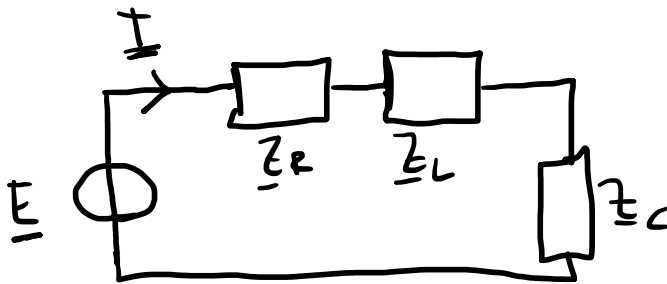
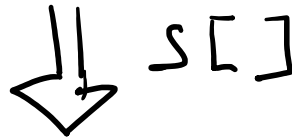
# RISONANZA NEI CIRCUITI IN REGIME AC

venerdì 29 aprile 2022 09:27



$$e(t) = \hat{V} \cos(\omega t + \alpha_v) \quad V$$

Studio in freq. della  
 $i(t)$



$$\begin{cases} Z_{eq} = Z_R + Z_L + Z_C \\ Z_R = R \\ Z_L = j\omega L \\ Z_C = -\frac{j}{\omega C} \end{cases}$$

$$\begin{aligned} \rightarrow Z_{eq} &= R + j\omega L - \frac{j}{\omega C} = R + j\left(\omega L - \frac{1}{\omega C}\right) = \\ &= R + jX \end{aligned}$$

Esiste un valore di  $\omega$  per il quale la  
reattanza si annulla:

$$\omega_0 L - \frac{1}{\omega_0 C} = 0 \rightarrow \omega_0^2 LC - 1 = 0$$

$$\rightarrow \omega_0^2 LC = 1 \rightarrow \omega_0^2 = \frac{1}{LC}$$

$$\boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$$

PULSAZIONE  
DI RISONANZA

$$\underline{I} = \frac{\underline{V}}{\underline{Z}_q} = I e^{j\alpha_I}$$

$$\underline{V} = V e^{j\alpha_V}$$

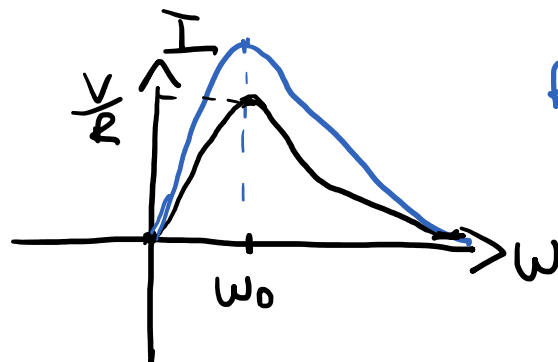
Supponiamo  $\alpha_V = 0 \rightarrow \underline{V} = V$

$$\underline{I} = I e^{j\alpha_I} \rightarrow |\underline{I}| = \frac{|\underline{V}|}{|\underline{Z}_q|} = \frac{V}{\sqrt{Z_{q,R}^2 + Z_{q,I}^2}} =$$

$$= \frac{V}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$

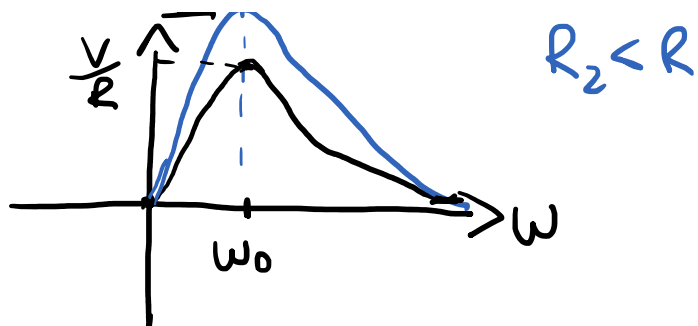
$$\rightarrow \alpha_I = -\varphi = -\arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)$$

| $\omega$   | $ \underline{I} $ |
|------------|-------------------|
| 0          | 0                 |
| $\omega_0$ | $\frac{V}{R}$     |
| $\infty$   | 0                 |



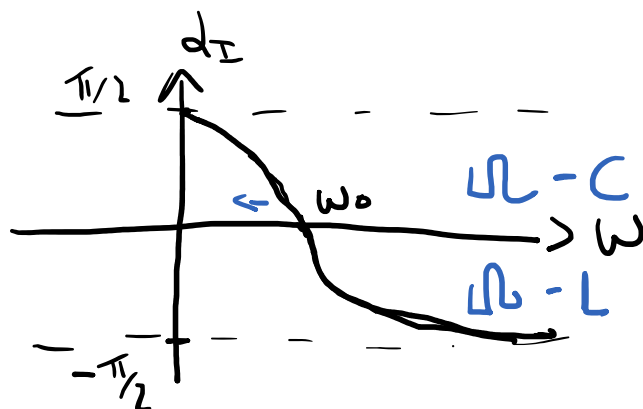
$R_2 < R$

| $\omega$   | $ I $         |
|------------|---------------|
| 0          | 0             |
| $\omega_0$ | $\frac{V}{R}$ |
| $\infty$   | 0             |



$$\varphi_I = \alpha_V - \varphi \rightarrow \alpha_I - \varphi = - \arctan \left( \frac{\omega L - \frac{1}{\omega C}}{R} \right)$$

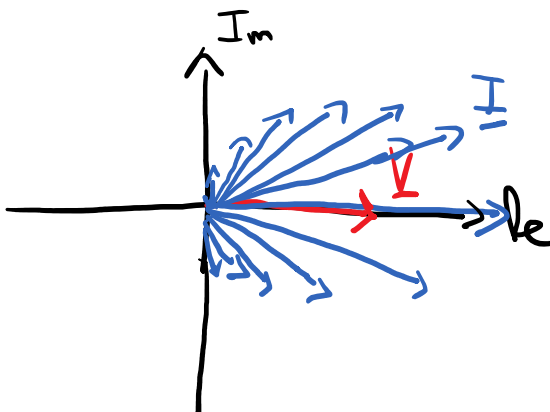
| $\omega$   | $\alpha_I$ |
|------------|------------|
| 0          | $\pi/2$    |
| $\omega_0$ | 0          |
| $\infty$   | $-\pi/2$   |



$$Z_{eq} = R + j(X_L - X_C)$$

$$\frac{V}{Z_{eq}} = \frac{V_R + V_L + V_C}{Z_{eq}}$$

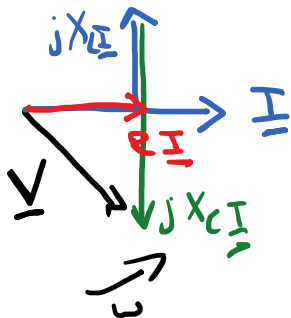
$$\omega_0 \rightarrow X_L = X_C$$



$$\omega < \omega_0 \rightarrow X_C > X_L$$

$$\underline{V} = \underline{V}_R + \underline{V}_L + \underline{V}_C$$

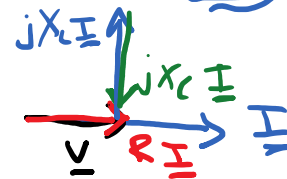
$$= \underline{R}\underline{I} + jX_L\underline{I} + jX_C\underline{I}$$



$$\omega = \omega_0 \rightarrow X_C = X_L$$

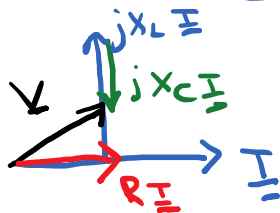
$$\underline{V} = \underline{V}_R + \underline{V}_L + \underline{V}_C$$

$$= \underline{R}\underline{I} + jX_L\underline{I} + jX_C\underline{I}$$



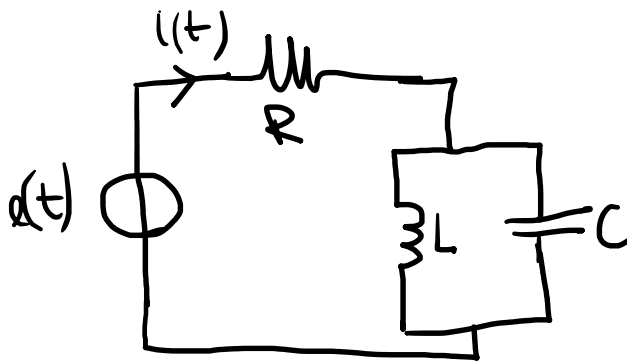
$$\omega > \omega_0 \rightarrow X_L > X_C$$

$$\underline{V} = \underline{R}\underline{I} + jX_L\underline{I} + jX_C\underline{I}$$

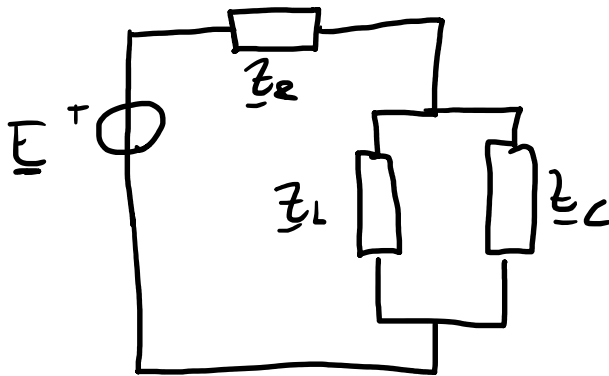


# ANTI RISONANZA

venerdì 29 aprile 2022 10:31



↓ S[]



$$\underline{Z}_{eq} = R + jX$$

$$v(t) = \hat{V} \cos(\omega t + \alpha_v) \text{ V}$$

$$\underline{Z}_{eq} = \underline{Z}_R + \underbrace{\underline{Z}_L // \underline{Z}_C}_{\underline{Z}_{LC}}$$

$$\underline{Z}_{LC} = \frac{\underline{Z}_L \cdot \underline{Z}_C}{\underline{Z}_L + \underline{Z}_C} = \frac{j\omega L \cdot \left(\frac{1}{j\omega C}\right)}{j\omega L + \frac{1}{j\omega C}}$$

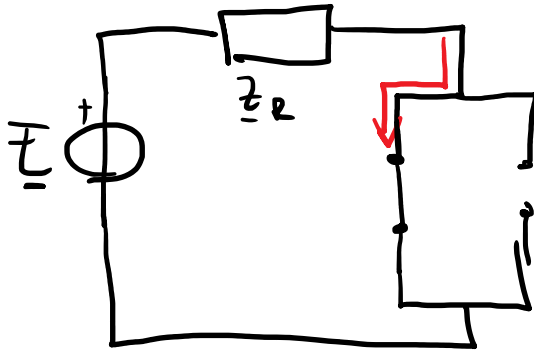
$$= \frac{L/C}{-\omega^2 LC + 1} \cdot j\omega L =$$

$$= j\omega L \frac{L/C}{1 - \omega^2 LC} = j \underbrace{\frac{\omega L}{1 - \omega^2 LC}}_X$$

Esiste una particolare  $\omega$  che rende la reattanza infinita:

$$1 - \omega^2 LC = 0 \rightarrow \boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$$

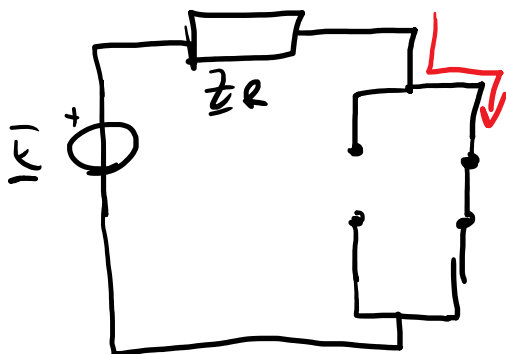
$$\omega = 0$$



$$X_L = 0$$

$$X_C = \infty$$

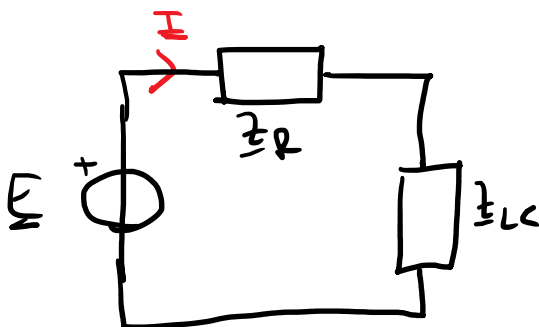
$$\omega \rightarrow \infty$$



$$X_L = \omega L \rightarrow \infty$$

$$X_C = -\frac{1}{\omega C} \rightarrow 0$$

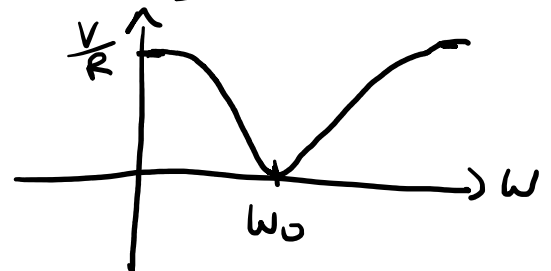
$$\omega = \omega_0$$



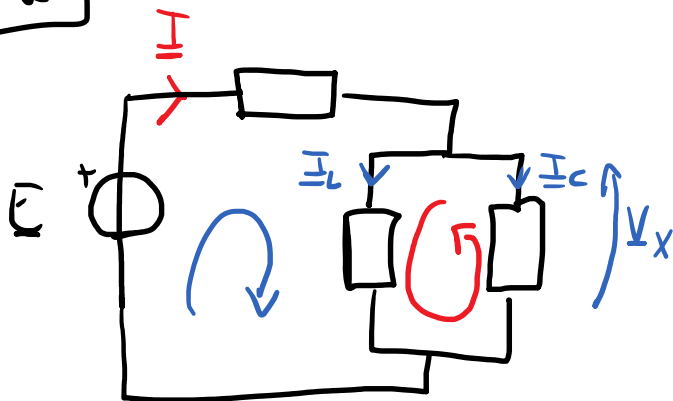
$$I = 0$$

$$X \rightarrow \infty$$

$$|I| = \frac{|V|}{|Z_T|} = \frac{V}{\sqrt{R^2 + X^2}}$$



$$\omega = \omega_0$$



$$\underline{I} = 0 \Rightarrow \underline{I}_L = 0$$

$$\underline{I}_C = 0$$

$$\underline{I} = \underline{I}_L + \underline{I}_C = 0$$

$$\Rightarrow \underline{I}_L = -\underline{I}_C$$

$$\underline{E} - \cancel{R\underline{I}} - \underline{V}_x = 0 \rightarrow \underline{V}_x = \underline{E}$$

$$\underline{I}_L = \frac{\underline{V}_x}{\underline{Z}_L} = \frac{\underline{E}}{j\omega L} = \frac{\underline{E}}{j\omega_0 L} = \frac{\underline{E}}{j\frac{1}{\sqrt{LC}}L} =$$

$$= -j \frac{\underline{E}}{\sqrt{\frac{L}{C}}} = -j \sqrt{\frac{C}{L}} \underline{E}$$

$$\underline{I}_C = \frac{\underline{V}_x}{\underline{Z}_C} = \frac{\underline{E}}{\frac{1}{j\omega C}} = j\omega_0 C \underline{E} = j \frac{1}{\sqrt{LC}} \cdot C \underline{E} =$$

$$= j \sqrt{\frac{C}{L}} \underline{E}$$

$$\underline{I}_L = -\underline{I}_C$$